

5/07/24

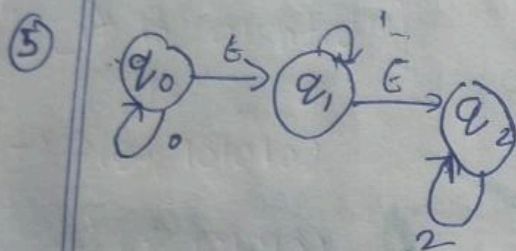
Assignment-2

① Construct NFA over $\{a, b\}$ that accept string a^*b as a substring.

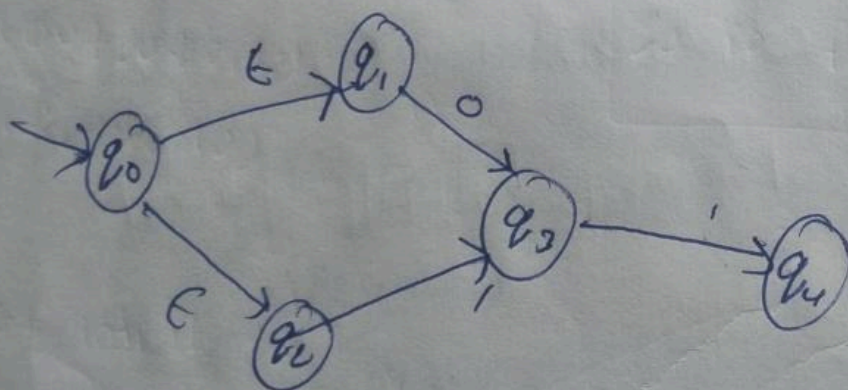
② $L = \{w \mid w \in a^*b^* \text{ and } |w| \geq 10\}$

③ Construct NFA that accept the language $L = \{a^n b^m\}$

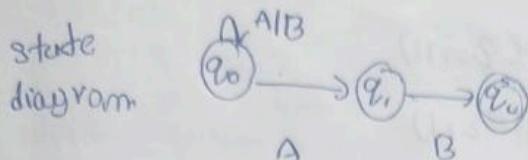
④ Consider following NFA



NFA $\rightarrow$ DFA



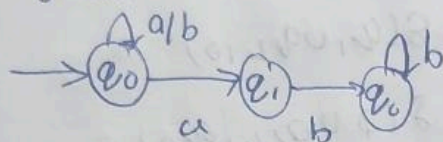
1) construct NFA over alphabet $\Sigma = \{a, b\}$ that accept string A^1B as substring



2) $L = \{w \mid w \in ab^*\}$ non empty

Alphabet $\Sigma = \{a, b\}$ $L = \{ab, aabb, \dots\}$

state diagram



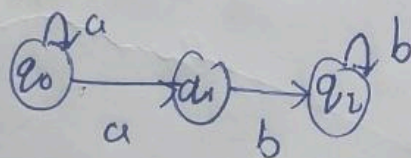
3) construct NFA that accept language

$L = \{a^n b^m \mid n \geq 1\}$

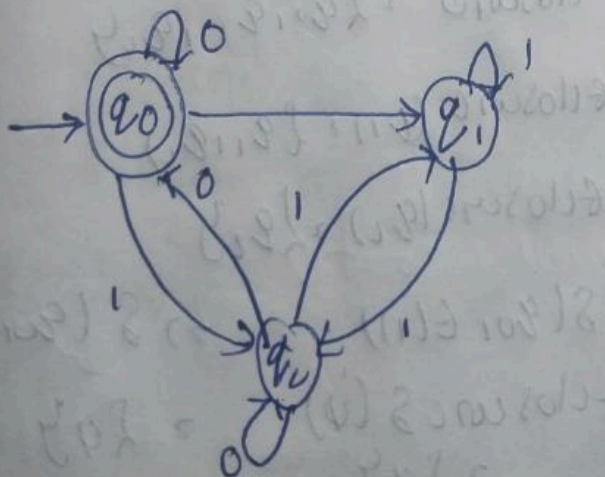
alphabet $\Sigma = \{a, b\}$

$L = \{a^1b^1, a^2b^2, a^3b^3, \dots\}$

state diagram



4) consider following NFA and check whether input 010-0011, 101010 is accepted or not



transition table

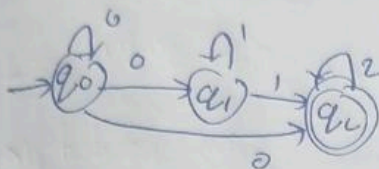
	0	1
q ₀	q ₀	q ₁
q ₁	∅	q ₁ , q ₂
q ₂	q ₁ , q ₀	q ₁

$\delta(q_0, \epsilon(0))$
 closure (0)
 closure 0

$\rightarrow \delta(q_0, \epsilon(1))$
 closure (1)
 $\{q_0, q_1\}$

$\rightarrow \delta(q_0, \epsilon(2))$
 closure (2)
 $\{q_0\}$

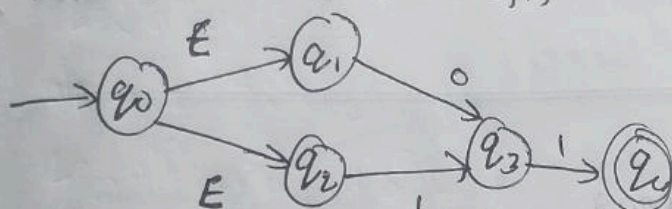
state diagram



transition table:

	0	1	2
$\rightarrow q_0$	q_0/q_1	ϕ	ϕ
q_1	ϕ	q_1, q_2	ϕ
$* q_2$	ϕ	ϕ	q_2

6 conversion of nfa to dfa



Table

	0	1
$\rightarrow q_0$	q_1	q_3
q_1	q_3	ϕ
q_2	ϕ	q_3
$* q_3$	ϕ	q_4
q_4	ϕ	ϕ

s-1: initial: q_0 final: q_4
 input $\{0, 1\}$

s-2: ϵ closure $q_0 = \{q_0, q_1, q_2, q_3\}$

ϵ closure $q_1 = \{q_1, q_3\}$

ϵ closure $q_2 = \{q_2, q_3\}$

ϵ closure $q_3 = \{q_3\}$

ϵ closure $q_4 = \{q_4\}$

ϵ closure $\{q_1, q_3, \phi\}$

ϵ closure $\{q_2, \phi\} \rightarrow \textcircled{B}$

$\rightarrow$ Transition table

$\delta(A, 0)$

$\rightarrow \epsilon$ closure $\{q_1, q_3, \phi\}$

ϵ closure $\{q_2, \phi\} \rightarrow \textcircled{B}$

s-3 $q_0 = \{q_0, q_1, q_2\}$ - initial state.

$\delta(A, 0)$

$\{q_0, q_1, q_2, \phi\}$